

What They Don't Tell You About Passive Investing

ACTIVE FUNDAMENTAL EQUITY | GLOBAL EMERGING MARKETS TEAM | INVESTMENT INSIGHT | 2017

The exodus from active to passive funds may be reaching bubble-like proportions, driven by an exaggerated critique of active management. In the United States, historically, many active managers have tended to outperform in bear markets, but underperform in bull markets.

Recently, the exodus has spread to active funds in the international (ex-U.S.) and emerging markets (EM) spaces. We attribute this spreading contagion to guilt by association, because we believe it has nothing to do with performance.

Over the past 15 years, most active international funds that are benchmarked against the MSCI All Country World Index ex-U.S. funds have outperformed their benchmark by 107 bps, net of fees, on a median basis, and most active EM funds that are benchmarked against the MSCI Emerging Markets Index have outperformed their benchmark by 76 bps, net of fees, on a median basis.¹ Over the same period, the median active international fund has outperformed the benchmark more than 70 percent of the time, while the median emerging market fund has outperformed more than 60 percent of the time.¹

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Throughout this document, performance data quoted is as of March 31, 2017 and is sourced from eVestment, unless otherwise noted. **Past performance is no guarantee of future results.** For methodology on calculations used in this document, please refer to the disclosures on the last page. Bear markets encompass periods of declining S&P 500 Index performance (2008); bull markets encompass periods of rising S&P 500 Index performance (2003 to 2007, and 2009 to present). "International" refers to non-U.S. equity funds, which are often benchmarked against the MSCI All Country World Index ex-U.S. "U.S." refers to U.S. equity funds which are often benchmarked against the S&P 500 Index. "Emerging Markets" refers to EM equity funds which are often benchmarked against the MSCI Emerging Markets Index.

¹ MSIM, eVestment, as of March 31, 2017.



Further, in emerging markets, most active managers beat the vehicle of choice for passive providers—the exchange-traded fund (ETF)—by an even wider margin. Whether you look at performance over 5 years, 10 years or since inception of the representative EM ETF, the median active EM manager outperformed the representative rival ETF by a margin of at least 115 bps, even after accounting for the fee differential (see *Display 9*).^{1,2}

The Great Exodus

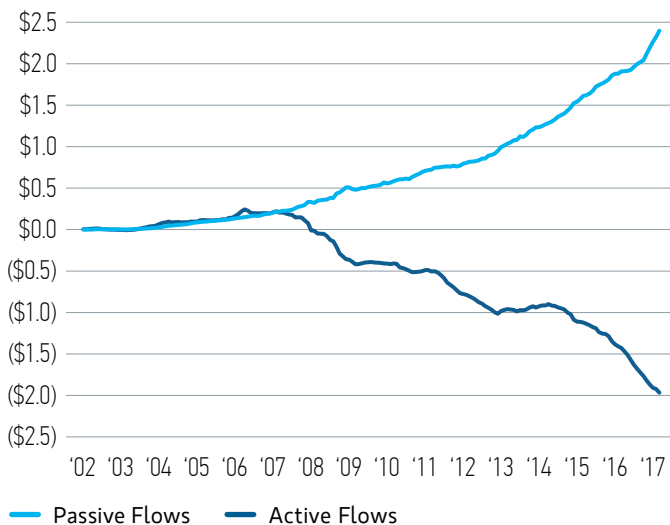
The gentle swell of equity flows into passive funds that began in the United States after the financial crisis of 2008 has given way to a worldwide tidal wave over the past two years, as *Display 1* illustrates. But there are reasons to expect this phenomenon could turn. For nearly three decades, or as far back as good data exists, most passive providers have outperformed most active managers in bull markets—particularly in the late stages of a bull market.¹ Flows often chase performance, and following a long streak of underperformance for U.S. large-cap managers, most investors are once again asking questions about why they should pay an active manager to pick stocks, when many passive providers are charging less and doing better.³

This trend of outperformance for many passive funds was evident in the second half of the 1980s, and again during the manic bull

DISPLAY 1

The Exodus from Active to Passive

Cumulative flows of equity funds since 2002 (US\$ trillions)

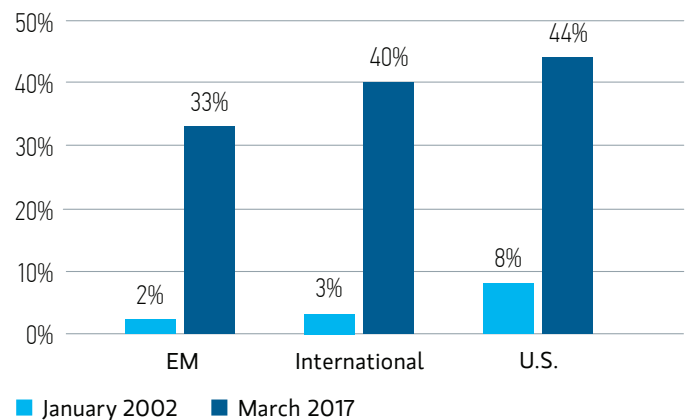


Source: EPFR Global, using cumulative flows from January 1, 2002 through March 31, 2017 of all equity funds in EPFR Global.

DISPLAY 2

The Rise of Passive

Passive share of total assets in managed equity funds



Source: EPFR Global, January 1, 2002 and March 31, 2017. Using passive funds referencing furthest historical data available in EPFR Global. EEM data includes all EM equity funds, Global Ex-US includes all Global Ex-US equity funds and USA includes all USA equity funds in EPFR Global.

market phase of the late 1990s. But flows towards passive products have exploded like never before during the current bull market, facilitated to a great degree by the rise of new passive investing options like the ETFs. Since 2002, passive equity funds (including U.S., international, EM and country-specific funds) have sucked in flows of approximately \$2.4 trillion, a sum that roughly matches the outflow from active funds of approximately \$2.0 trillion.⁴

The exodus from active to passive is now going parabolic. Over the last year, investors withdrew \$538 billion from active funds worldwide, and poured \$489 billion into passive funds.⁵ Measured as a percentage of total equity market assets, this surge is the largest on record. As shown in *Display 2*, since 2002, passive funds barely registered as a share of total equity market assets in each respective universe, but as of March 2017, they account for a sizable 44 percent share in the United States, 40 percent in international, and 33 percent in emerging markets.⁶

We believe the flow of money into passive funds is exhibiting the classic characteristics of a bubble, beginning with near universal excitement for a big idea—in this case, the notion that cheap index funds are the wave of the automated investing future. During bubble periods, the excitement built around the big idea generates an explosion of interest in not only the main product but in ancillary products. In the current instance, there has also been a surge in the indexes the ETFs track. As shown in

² Please refer to the source description for Display 9 for more detailed methodology regarding ETF versus active EM fund performance.

³ eVestment, Bloomberg, as of March 31, 2017.

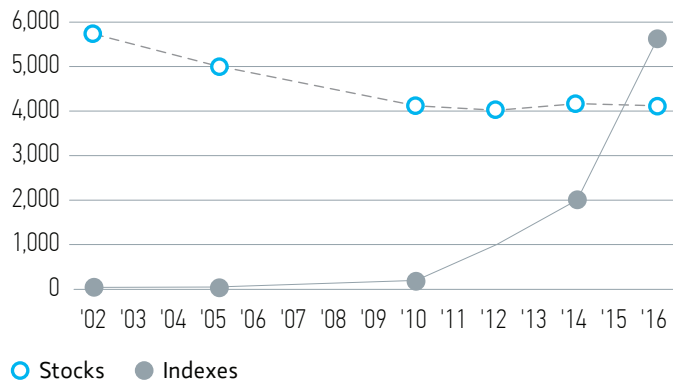
⁴ EPFR Global, using cumulative flows from January 1, 2002 through March 31, 2017.

⁵ EPFR Global, using cumulative flows from March 1, 2016 through March 31, 2017.

⁶ EPFR Global, January 1, 2002 and March 31, 2017. Using passive funds referencing furthest historical data available in EPFR Global. EEM data includes all EM equity funds, Global Ex-US includes all Global Ex-US equity funds and USA includes all USA equity funds in EPFR Global.

DISPLAY 3**A Passive Bubble?**

More indexes than stocks in the U.S.



Data: Bloomberg Businessweek, Bloomberg Intelligence, Sanford C. Bernstein. Data as of 2016.

Display 3, 15 years ago in the United States, there were very few benchmark indexes; today, there are more than 5,000 market indexes, which is more than the number of publicly listed U.S. stocks.⁷ We have recently seen ETFs launched to track the ETF industry—a clear sign of euphoria over passive investing.

Unfortunately, investors fleeing active U.S. funds may be mistaking a cyclical trend for a structural shift. Meaningful performance data for active funds in the U.S. goes back to 1990, and reveals four distinct cycles. As shown in *Display 4*, in the U.S., many active managers outperformed the S&P 500 Index in the years from 1990 to 1994, underperformed from 1995 to 1998, and outperformed again for a long stretch spanning 1999 to 2010 before suffering another period of sustained underperformance from 2011 to 2016.¹

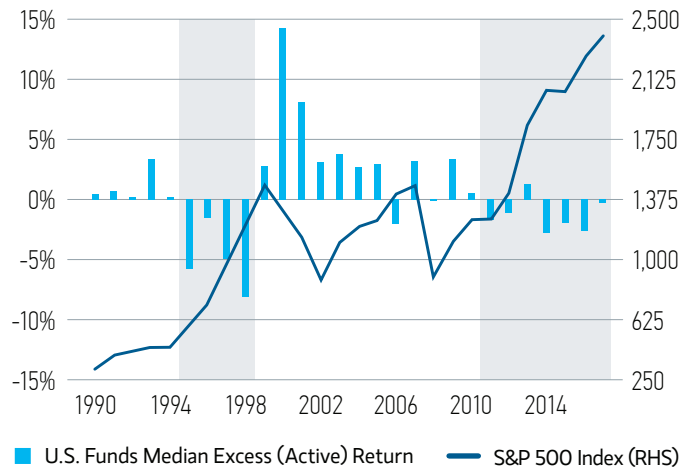
In short, many active managers in the United States have tended to underperform in bull markets, and outperform in bear markets, as illustrated in *Display 4*. If this pattern continues, the next turn in the cycle could bring a period of sustained active outperformance, just as investors are fleeing active funds en masse.

This year the U.S. bull market is nine years old. When the market turns, we believe the relative performance of many active managers is likely to stage a strong comeback.

Other researchers have traced the general pattern—that many active U.S. equity managers tend to underperform in bull markets and outperform in bear markets—much farther back in time. A 2006 study by Robert Kosowski of Imperial

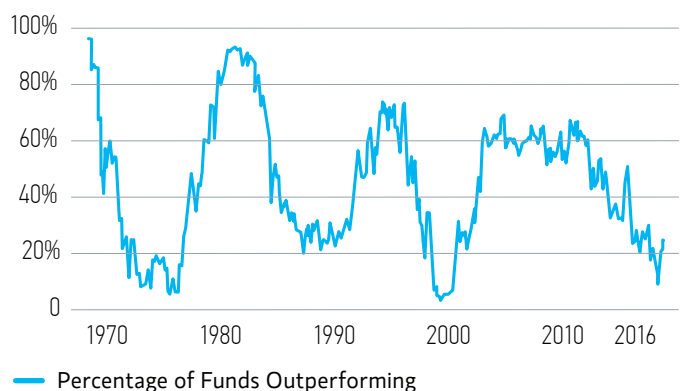
DISPLAY 4**Investors May Be Mistaking a Cyclical Trend for a Structural Shift⁸**

In the U.S., active performance tends to improve when the S&P 500 Index declines



Source: MSIM, eVestment, Bloomberg. Data as of March 31, 2017.

College, London, showed that from 1962 to 2005, active funds outperformed by 400 bps during recession periods, and underperformed by 133 bps in expansionary periods. Research by Minack Advisors confirms the basic pattern based on data going back to 1970 (*see Display 5*).

DISPLAY 5**Active Management Goes Through Cycles⁹**

Source: CRSP, Bloomberg, Robert Shiller, Instinet research, Marketwatch, Minack Advisors.

⁷ Bloomberg Businessweek, Bloomberg Intelligence, Sanford C. Bernstein. Data as of 2016.

⁸ Universe is U.S. funds in eVestment. U.S. funds median excess returns are net of fees and relative to the fund managers' preferred benchmark. The median U.S. funds returns are used each given year. Number of U.S. funds vary by year; averaged 178 funds per year since 1990.

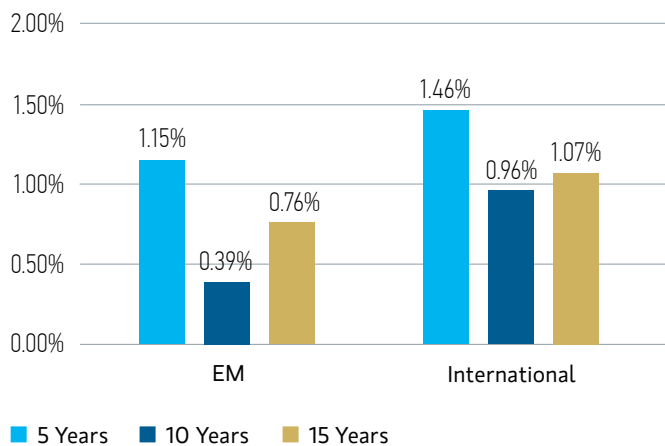
⁹ Data analysis from January 1970 to December 2016 and includes U.S. active equity mutual funds, outperforming the total return of the S&P 500 Index based on trailing five-year performance net of fees. Only includes those funds with five or more years of performance data and those that are classified as either growth or income.

There are a variety of explanations for why, as a group, active managers in the United States tend to underperform in bull markets. Particularly in the late stages of a bull market, dispersion of returns tends to be low and correlation of returns tends to be high—making it difficult for active managers to bring unique insights into the investment process. This challenge has been magnified in the bull market of the post-crisis era following 2008, because extremely loose monetary policy has further increased correlations, and pushed volatility to historic lows.¹⁰

As central banks move to normalize monetary policy, however, this should reduce the high correlation between asset classes, potentially giving active managers much greater opportunity to outperform. Since central banks are already moving in this direction and correlations between stocks and between different asset classes have finally begun to fall, we believe investors would be wise to reconsider active investing now, rather than wait for the next bear market to begin.

DISPLAY 6

Outside the U.S., Most Active Managers Outperformed Across Various Time Periods¹²



Source: MSIM, eVestment, as of March 31, 2017. Median fund is used from the EM and international active manager universes in eVestment. Outperformance is excess returns, net of fees and relative to the manager's preferred benchmark.

Guilty by Association: Active Managers in International and Emerging Markets

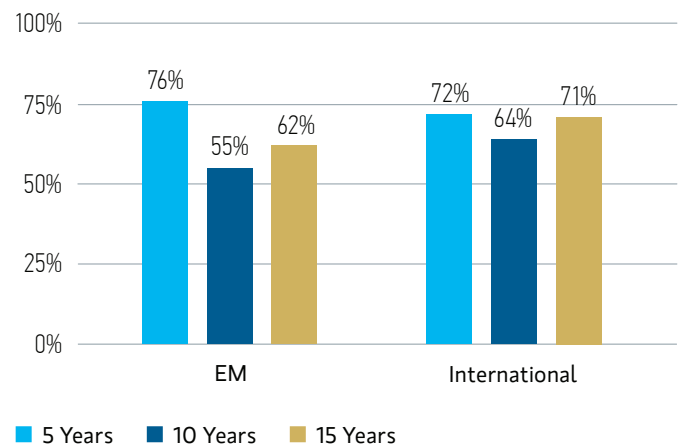
The strange trend of this bull market has been that, for no good reason, the popularity of passive investing hopped the fence and started to hit international and emerging markets hard. We explain this spreading contagion as guilt by association, because we believe it has nothing to do with performance. Active managers have delivered excess returns in international and emerging market equities, over various time periods.¹¹ For these markets, solid data goes back 15 years, and since then, most active managers have delivered outperformance—relative to the benchmark and net of fees—in both bull markets and bear markets.¹

Over the last five years, active international funds have outperformed their benchmark by 146 bps on a median basis, and active EM funds have outperformed by 115 bps on a median basis, even after accounting for fees (see Display 6).¹ As depicted in Display 7, over the same period, more than 70 percent of active international funds and active EM funds have outperformed their benchmark.¹

DISPLAY 7

In EM and International, the Majority of Active Funds Outperform¹³

Share of active funds outperforming



Source: MSIM, eVestment as of March 31, 2017. Median fund is used from the EM and international active manager universes in eVestment. Outperformance is excess returns, net of fees and relative to the manager's preferred benchmark.

¹⁰ MSIM, Bloomberg, as of March 31, 2017. Average pairwise correlation represents the average of 3-year weekly return correlations of all bivariate stock combinations at each point in time for S&P 500 constituents.

¹¹ Various time periods refers to the five and ten year periods leading up to March 31, 2017.

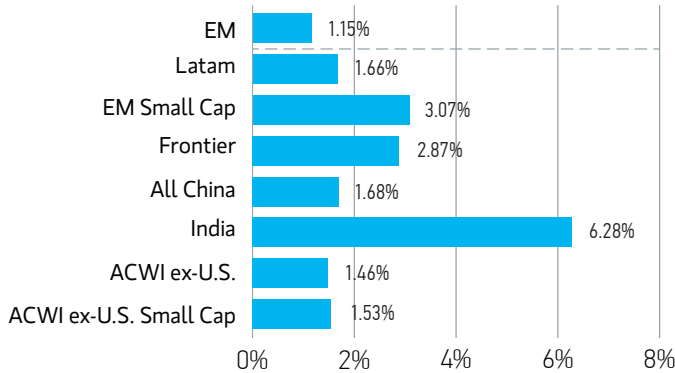
¹² For the five- and 10-year periods, only funds with data the entire periods are included. Calculated annualized returns for any fund with data from 2002-2017 and then calculated the median fund annualized returns. For the 15 year period, funds with data for any time period were included to prevent survivorship bias skewing results. Five-year (March 31, 2012 to March 31, 2017) period includes:

91 EM funds and 71 international funds. Ten-year (March 31, 2007 to March 31, 2017) period includes: 140 EM funds, 47 international funds. Fifteen-year (March 31, 2002 to March 31, 2017) period includes: 176 EM funds and 112 international funds.

¹³ For the five- and ten-year periods, only funds with data the entire periods are included. For the 15-year period, funds with data for any time period were included to prevent survivorship bias skewing results. Five-year (March 31, 2012 to March 31, 2017) period includes: 91 EM funds and 71 international funds. Ten-year (March 31, 2007 to March 31, 2017) period includes: 40 EM funds and 47 international funds. Fifteen-year (March 31, 2002 to March 31, 2017) period includes: 176 EM funds and 112 international Funds.

DISPLAY 8**Most Specialized Funds Outperform¹⁴**

Active performance by EM and international equity classes over the last five years



Source: MSIM, eVestment, data from March 31, 2012 to March 31, 2017. Calculations as excess returns which are net of fees and relative to the manager's preferred benchmark.

The longer-term picture—the last 10 or 15 years—is the same. Looking at the past 15 years, for example, active international funds have outperformed by 107 bps, and active emerging market funds have outperformed by 76 bps.¹ Similarly, over the same time period, 71 percent of active international funds and 62 percent of active emerging market funds have outperformed their benchmark.¹

Not only have most active international and emerging market funds outperformed over the same period, but, as a group, they have rarely suffered bad years. The median active international manager has underperformed in five of the last 15 years and the median active EM manager has underperformed in just three of those years.^{1, 15}

The Upside of Specialization

We think one reason why active managers regularly outperform in international and emerging market equities is increased specialization. Even in periods when interest rates and correlations are high, there is much greater opportunity to pick winners in niche markets with a multitude of small, often poorly researched and understood players.

In emerging markets and international as well, funds that are more specialized by geography (region, country, frontier) or asset

class (small caps) tend to outperform their respective indexes by a bigger margin than less specialized funds, as shown in *Display 8*. When global funds enter emerging markets, they tend to buy just the largest companies in the index, leaving alpha on the table for local funds. This is also true over various time periods.^{1, 16}

The Passive Vehicle of Choice Is a Clunker

The case for active managers is even stronger when one compares their performance not against the hypothetical returns of an index, but against the actual returns of passive providers. The investment vehicle of choice for passive providers is the ETF, which is exploding in popularity. Exchange-traded funds were specks on the sidelines of the markets as late as 2000, but last month smashed past \$4 trillion in assets. That includes roughly \$2.8 trillion in the United States, with the rest in international and emerging markets.¹⁷

In the United States, the major equity market ETFs perform more or less as advertised. They charge minimal fees, track their target benchmark indexes quite closely.

However, the same cannot be said of international and emerging markets ETFs, which tend to have much higher expenses and underperform their underlying indexes. In fact in EM, the representative ETF has significantly underperformed the rival active manager by a margin of at least 115 bps, even after accounting for the fee differential, over various time frames.^{1, 2} In international markets, the same basic pattern holds—ETFs have underperformed active funds, by a significant margin.^{1, 2} *Display 9* depicts this surprising story.

Thanks to the power of compounding, these performance advantages could amount to a significant sum over time: For example, *Display 10* shows how the five-year performance advantages depicted in *Display 9* could pay off, assuming \$100 invested today for five years, and annual returns averaging 6 percent in EM and 5 percent in international markets.

Why Active Beats Passive in Emerging Markets and International

Passive providers do not, and in fact cannot, provide many services that we believe are particularly critical to performance in international markets. Because passive funds simply track emerging market indexes, they have no choice but to invest heavily in large-cap stocks, which in many EMs include vast, inefficient state companies that are not governed to maximize profits or respect the rights of minority shareholders.

¹⁴ Calculated annualized returns for funds with data from 2012 - 2017 and then calculated the median fund's annualized returns. Number of funds used: EM: 176, Latam: 19, EM small cap: 37, Frontier: 20, All China: 42, India: 7, ACWI ex-U.S.: 71, ACWI ex-U.S. Small Cap: 22.

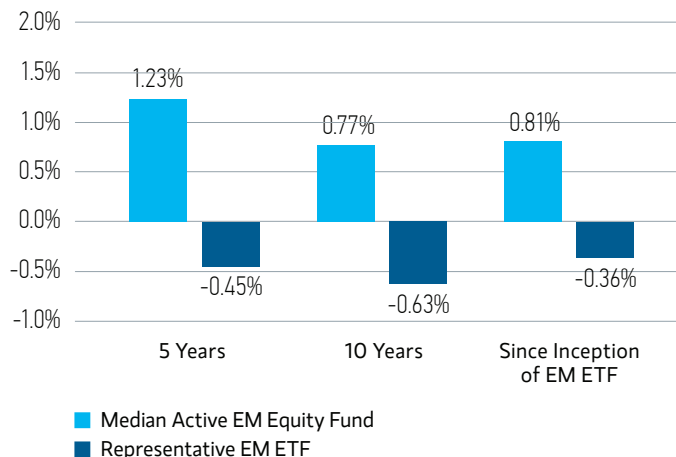
¹⁵ Number of funds vary by year but over the period, averaged 87 funds for EM and 264 funds for the U.S.

¹⁶ We also analyzed specialized funds' active performance over 10 and 15 years and found the results to be largely in line with the five year performance.

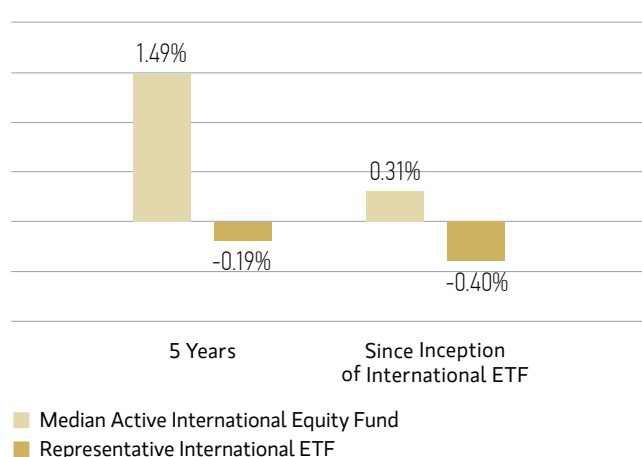
¹⁷ ETFGI, as of March 31, 2017

DISPLAY 9**Active Management Tends to Beat ETFs¹⁸
In Emerging Markets**

Excess returns against MSCI Emerging Markets Net Index

**And in International Markets**

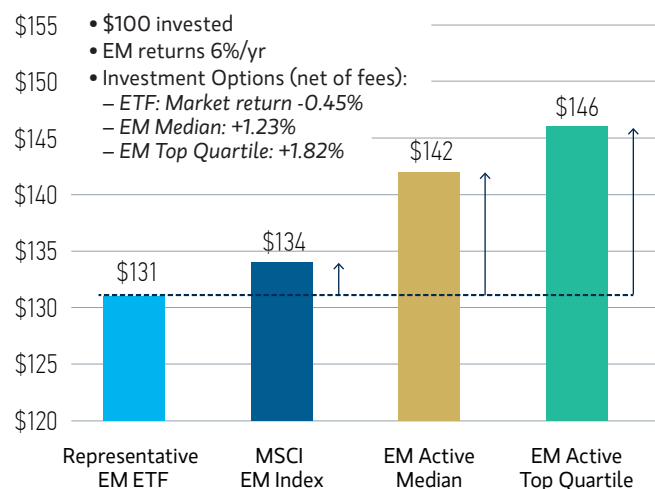
Excess returns against MSCI All-Country World Index ex-U.S.



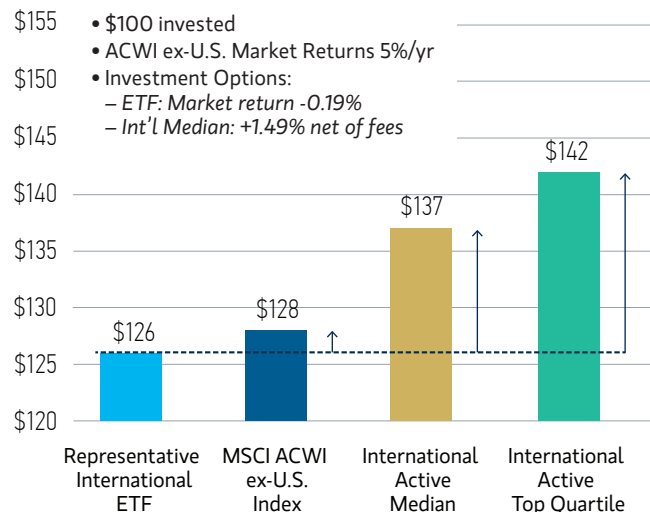
Source: Bloomberg, eVestment, as of March 31, 2017. Excess returns are net of fees, relative to the benchmark. Representative ETFs are the largest ETFs in their respective universes.

DISPLAY 10**How Active Management Could Pay Off in EM¹⁹**

Five-year compounded returns of \$100

**How Active Management Could Pay in International Markets¹⁹**

Five-year compounded returns of \$100



Source: MSIM, eVestment, from March 31, 2012 to March 31, 2017.

¹⁸ Representative EM ETF: iShares Emerging Markets Equity ETF (EEM), inception date is April 2003, benchmarked to MSCI EM Index. Representative international ETF: iShares ACWX, inception date is March 2008, benchmarked to MSCI ACWI ex-U.S. Since inception returns include only funds in each universe that had data for each year. Median active EM equity fund is median of funds in eVestment's Active EM Equity universe that are benchmarked to MSCI EM Index Net Dividends or MSCI EM Index Gross Dividends. Number of funds were 79, 35 and 23 for the five year, 10 year and since inception periods. Median active international fund is median of funds in eVestment's All-Country World Index

ex-U.S. All Cap Equity universe that are benchmarked to MSCI ACWI ex-U.S. Number of funds were 55 and 37 for the five year and since inception periods.

¹⁹ Assume 600 bps annualized for EM index and 500 bps annualized for MSCI ACWI ex-U.S. based on MSIM GMA estimates. Assume -45 bps, +123 bps, +192 bps for representative EM ETF, EM Active Median and EM top quartile respectively based on empirical findings. Assume -19 bps, +149 bps, +227 bps for MSCI ACWI ex-U.S. ETF, International Active Median and International top quartile respectively based on empirical findings.

There are thousands of stocks in the emerging market universe, but less than 1,000 are included in the MSCI Emerging Markets Index.²⁰ Unlike passive index funds, active funds can invest in non-index stocks in an effort to boost returns. They have the time and resources to do due diligence on stocks, which is particularly critical in often opaque emerging markets. Active managers seek to maximize returns by hedging for currency movements, which play a big role in dollar returns in EM—historically accounting for a third of returns.²¹ In rough international waters, we believe you are better off with an active captain.

Finally, we believe passive providers often do not perform as advertised. The basic idea of passive is, well, very passive: If your active manager can't beat the market, why not settle for a cheap ETF to at least keep pace with it? But these ETFs are often not as cheap or risk free as many people assume.

If an ETF charges a fee of 1 percent to track an index, then, in theory, its returns will lag the index by 1 percent—which is why ETFs compete so hard to lower fees. However, many other costs can cut into ETF returns, including the transactions costs involved in buying or selling individual stocks, or rebalancing the portfolio, and cash drag.²²

It is too early to say for sure, but there is some evidence that investors are starting to understand these trends and reverse their judgment on active managers, at least in emerging markets: Strong performance in April 2017 just turned the flow into active EM funds positive for the year, with a total inflow of around \$2.5 billion.²³

Summary

The popularity of passive investing appears to be reaching the late stages of a market cycle, and may be poised for a sharp turn. When people start allocating money to a fad in complete disregard of performance fundamentals, you know the trend is nearing exhaustion.

We believe the now-widespread assumption that active managers only extract fees and underperform over time is false—the idea

that active managers underperform in all markets is built on a misleading critique of their performance in the United States. In the U.S., the average active manager has outperformed in bear markets, but underperformed in bull markets.¹ In the current bull market, easy monetary policy has exaggerated this challenge, narrowing correlations and all but erasing volatility, leaving active managers little room for differentiating between stocks. But that challenge will likely drop sharply as central banks move to normalize monetary policy, a process that has already begun.

More significantly, our research shows that the more specialized the asset class, the more active managers tend to outperform. This is particularly true of most active managers in international and emerging equity markets: They tend to outperform their relevant benchmark, after fees, and over various time frames. Even more striking, most have outperformed the vehicle of choice of passive providers—the ETF.^{1,3}

We believe this is a bad time to join the crowd in moving from active to passive investing in the United States, and there has never been a good time to make that move in international or emerging markets.

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²⁰ EMSCI, Bloomberg, as of March 31, 2017

²¹ MSIM calculations, as of December 2016.

²² Cash drag: When a lot of money is flowing into a fund, particularly late in a bull market, managers often cannot deploy it fast enough, so cash grows as a share of the portfolio and drags down returns.

²³ EPFR Global, as of April 2017.

Methodology for calculations used: EM, U.S. and international active manager universes are based on eVestment categorizations, and screened for long-only actively managed funds. EM active managers include those in the Emerging Markets All Cap Equity universe (192 funds, covering \$403.9 B in assets). U.S. active managers include those in the US All Cap Equity universe (434 funds, covering \$855.7 B in assets). International active managers include those in the All Country World Index ex-U.S. All Cap Equity universe (113 funds, covering \$426.9 B in assets). Performance data quoted is annualized and net of fees. eVestment reporting is done by fund, not by manager. eVestment relies on manager submissions of data. There is the potential for selection bias based on this structure and also the possibility of survivorship bias. Inactive funds are included in our study. We compared all funds on a net basis for fair comparison with ETFs. We also compared all net excess returns according to the manager's specified benchmark for each fund. **Past performance is no guarantee of future results.**

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The **MSCI Emerging Markets Index (MSCI EM)** is a free float-adjusted market capitalization weighted index that is designed to measure equity market performance of emerging markets.

The **MSCI All Country World Ex-U.S. Index** is a free float-adjusted market capitalization weighted index designed to measure the equity market performance of developed and emerging markets, excluding the U.S. The term "free float" represents the portion of shares outstanding that are deemed to be available for purchase in the public equity markets by investors. The performance of the Index is listed in U.S. dollars and assumes reinvestment of net dividends.

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